

Coding & Solution

Date	06 July 2026
Team ID	xxxxxx
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	[https://github.com/Harshithamakina/SmartBridge-Voice_Based_Concept_Understanding_Analyser]
Programming Language(s)	Python
Framework(s) Used	Streamlit, FastAPI
Key Features Implemented	Audio Upload, Speech-to-Text, Semantic Analysis, Audio Analysis, PDF Report
Pending / Incomplete Features	No
Setup / Run Instructions	Install dependencies → Run <code>streamlit run app.py</code> → Upload audio → Analyze

Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The application follows a modular architecture with separate components for speech transcription, semantic analysis, audio processing, scoring, and report generation. The code is maintained using GitHub for version control and is designed for future scalability and enhancements.